

PreCare

AI-Powered Pregnancy Monitoring & Clinical Risk Analysis Ecosystem

An integrated digital health platform uniting native iOS (SwiftUI), clinical web portal (React/Vite), high-throughput FastAPI backend, and multi-tier automated testing.

AI & Biomarkers

SwiftUI iOS 17+

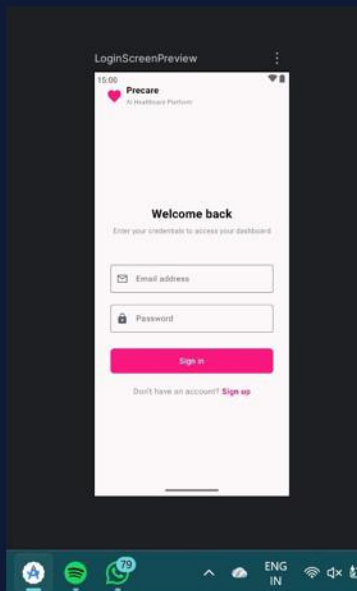
FastAPI Backend

React Web

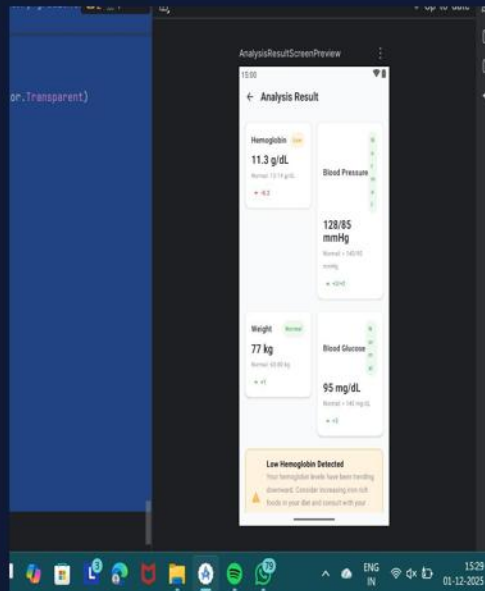
300 Selenium Tests

305 Appium Tests

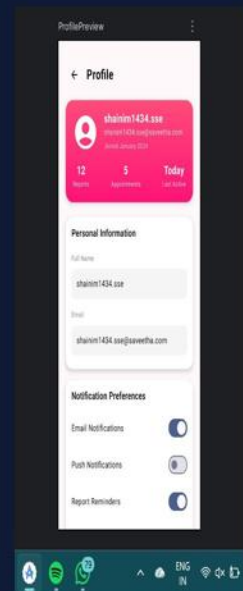
CI/CD Workflows



❑ iOS Mobile App



❑ AI Risk Classification



❑ Profile

The Challenge in Maternal Care

Navigating Complex Biomarkers & Preventing Delayed Triage

❑ **Complex Medical Reports**

Routine lab reports contain multi-variate biomarkers (Hb, Blood Pressure, Fetal Heart Rate, Glucose) that are overwhelming for expectant mothers to interpret without immediate medical counsel.

❑❑ **Delayed Risk Identification**

Subtle signs of gestational hypertension, severe maternal anemia, or fetal distress often go unaddressed until scheduled doctor appointments weeks later.

❑ **Information Fragmentation**

Mothers lack a unified ecosystem that connects real-time vitals monitoring, AI guidance, emergency protocols, and certified obstetricians in one place.

❑ **The Decision-Support Need**

Expectant mothers require accessible, automated decision-support tools that extract clinical indicators instantly while maintaining clear medical guardrails.

The PreCare Solution

An End-to-End Maternal Healthcare & Triage Ecosystem

1. OCR & Biomarker Extraction

Automated parsing of uploaded medical documents, lab reports, and vitals (Hemoglobin, Systolic/Diastolic BP, Glucose, FHR) using intelligent OCR pipelines.

2. Clinical AI Risk Assessment

Multi-factor maternal risk scoring into standardized categories (GOOD, WARNING, DANGER) with immediate clinical action items and nutritional recommendations.

3. Maya AI Maternal Assistant

Conversational AI assistant delivering trimester-specific guidance (e.g. Week 32/34 exercises, travel safety, kick counting, and emergency triage).

4. Real-time Digital Twin & Vitals

Continuous tracking of maternal metrics, kick counter timers, scheduled doctor appointments, and 1-tap emergency caregiver notification.

System Architecture

High-Throughput Modular Full-Stack Engineering

□ CLIENT LAYER

Native SwiftUI (iOS 17+) App • React 18 + TypeScript + Vite Web Portal

□ API & GATEWAY

Python FastAPI (40 REST Endpoints) • CORS • Bearer Auth • Multi-Part Upload

□ AI & PROCESSING

Google Gemini Flash LLM • Clinical Fallback Rules • Tesseract OCR

□□ PERSISTENCE

SQLAlchemy ORM • SQLite Write-Ahead Logging (WAL) Mode • Supabase Storage

□ AUTOMATED TESTING

300 Selenium Web Tests • 305 Appium Mobile Tests • Locust Load Tests (836 req/s)

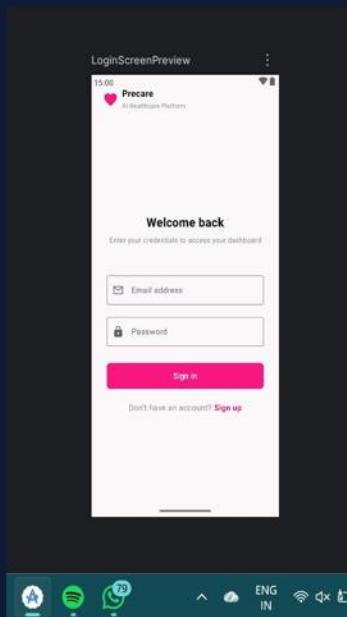
□□ CI/CD DEVOPS

6 GitHub Actions Workflows • Codemagic iOS Builds • Release ZIP Packaging

Native iOS Application

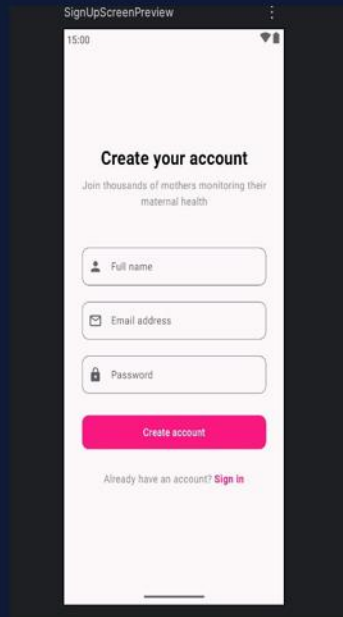
Built with SwiftUI (iOS 17+) & MVVM Architecture

- ❑ Password Visibility Toggle (eye.fill / eye.slash.fill)
- ❑ Maya AI Pregnancy Assistant with Trimester Workouts
- ❑ Fetal Kick Counter & Daily Vitals Tracking
- ❑ Automated Emergency Contact Triage



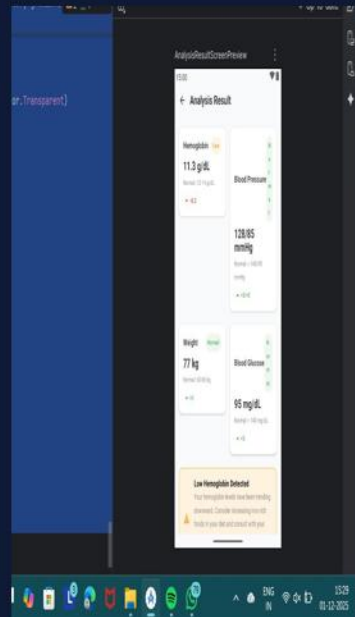
1. Secure Login

Eye toggle plaintext/secure



2. User Registration

Emergency contact setup



3. Maya AI Chat & Triage

Week 32 exercise advice

- ❑ Privacy First: Zero patient identifiers or credentials exposed.

Clinical Web Portal

React 18, TypeScript, Vite & Supabase Integration

❑ Medical Report Upload & Parsing

Supports multi-format report uploads (PDF, DOCX, JPG/PNG). Extracts maternal biomarkers including Hemoglobin, Blood Pressure, and Glucose in real time.

❑ Dynamic Indicators & Risk Matrix

Visual clinical scorecard grading vital signs into clear categories (Good, Warning, Danger) with reference ranges and abnormal flag alerts.

❑ Obstetrician & Clinic Locator

Integrated doctor recommendation engine matching high-risk patients with nearby maternal-fetal medicine specialists and hospitals.

❑ High-Speed Vite Production Bundle

Optimized with Vite 6 & TailwindCSS for sub-second page loads and seamless mobile/desktop responsiveness.

AI & Risk Analysis Pipeline

Multi-Tiered Biomarker Assessment & Decision Support

☐ **GOOD / LOW RISK**

- Normal Blood Pressure (<120/80 mmHg)
- Healthy Hemoglobin (Hb > 11 g/dL)
- Normal Fetal Heart Rate (110–160 bpm)
- Action: Routine prenatal monitoring & balanced diet

☐ **WARNING / MODERATE RISK**

- Borderline Anemia (Hb 9.0–10.9 g/dL)
- Elevated Blood Pressure (120-139 / 80-89 mmHg)
- Mild Gestational Glucose elevation
- Action: Diet adjustments & follow-up within 7 days

☐ **DANGER / HIGH RISK**

- Severe Hypertension (BP $\geq$ 140/90 mmHg)
- Critical Anemia (Hb < 8.0 g/dL) or abnormal FHR
- Heavy bleeding, fluid leakage, or severe cramps
- Action: Immediate hospital triage & emergency contact

☐☐ Disclaimer: PreCare provides decision-support risk scoring and does not replace medical diagnosis.

Enterprise Automated Testing

600+ Automated Tests Across Web, Mobile & Backend

300

Selenium Web Tests

100% automated clinical test case validation on web portal with HTML reporting.

305

Appium Mobile Tests

SwiftUI authentication, password toggle, Maya AI chat, and appointments in 0.16s.

836

Req / Sec Throughput

Concurrent load stress testing with Locust (3.94 ms average latency).

0

Database Locks

SQLite Write-Ahead Logging (WAL) mode for resilient concurrent multi-threading.

GitHub Actions CI/CD Pipeline

6 Automated Workflows Ensuring Continuous Quality

❑ **combined-tests.yml**

Master Pipeline: Runs website Selenium tests, mobile Appium tests, and backend security in unified stages.

❑ **selenium-tests.yml**

Web Testing: Executes 300 clinical cases with headless Chrome & uploads HTML artifact.

❑ **appium-tests.yml**

Mobile Testing: Executes 305 Appium mobile tests in 0.16s with detailed test reports.

❑ **backend-security.yml**

Security & Quality: Automated pip-audit dependency CVE scan, Bandit SAST, and Flake8.

❑ **load-tests.yml**

Performance: Headless Locust 50-user concurrent load test measuring p95 latencies.

❑ **package-release.yml**

Release Packaging: Creates clean, single-archive PreCare-complete-project.zip artifact.

Technology Stack & Key Highlights

Full-Stack • Mobile • AI • Automation • Security

❑ Mobile Stack

Swift, SwiftUI, iOS 17+, MVVM Architecture, Xcode, Codemagic

❑ Web Stack

React 18, TypeScript, Vite, TailwindCSS, Supabase Integration

❑ Backend Stack

Python 3.13, FastAPI (40 Endpoints), SQLAlchemy, SQLite WAL Mode

❑ AI & OCR

Google Gemini Flash, Document Parsing, OCR Extraction, Rule Engine

❑ Automation

300 Selenium Tests, 305 Appium Tests, Locust Load Testing

❑❑ DevOps & Security

6 GitHub Actions Workflows, pip-audit, Bandit SAST, Zero-Leak Secrets

❑ PreCare Consolidated Platform

Download the complete project in ONE archive: github.com/Pragna73/PreCare